MOTOKI KAGAMI

Nara Institute of Science and Technology (NAIST) | Ikoma, Nara, Japan
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RESEARCH INTERESTS

Human-computer interaction; 3D user interfaces; adaptive and intelligent interfaces; VR/XR interaction; haptics and pseudo-haptics.

EDUCATION

Nara Institute of Science and Technology Apr 2026 - Present
Ph.D. in Engineering, **in progress** | Advisor: Prof. Kiyoshi Kiyokawa

Nara Institute of Science and Technology Apr 2024 - Mar 2026
M.S. in Engineering | Advisor: Prof. Kiyoshi Kiyokawa

The University of Aizu Apr 2020 - Mar 2024
B.S. in Computer Science | Advisor: Prof. Lei Jing

PEER-REVIEWED PUBLICATIONS

[1] **Motoki Kagami**, Yuta Kataoka, Yutaro Hirao, Monica Perusquia-Hernandez, Satoshi Hashiguchi, Hideaki Uchiyama, Kiyoshi Kiyokawa, Shohei Mori. "Tag-Along Virtual Windows Increase Perceived Resistance and Task Load in Augmented Reality." *IEEE Transactions on Visualization and Computer Graphics*, 32(5), 4344-4353, May 2026 (IEEE VR 2026 Special Issue). DOI: 10.1109/TVCG.2026.3679137 | Project page
Oral presentation by Motoki Kagami at IEEE VR 2026.

[2] **Motoki Kagami**, Zeping Yu, Sim Teck Ceng, Lei Jing. "Sign Language Vocabulary Recognition Only with Tactile Sensing Glove." *2024 IEEE 17th International Symposium on Embedded Multicore/Many-core Systems-on-Chip (MCSoC)*, 110-116, December 2024. DOI: 10.1109/MCSoC64144.2024.00029
Oral presentation by Motoki Kagami at MCSoC 2024.

RESEARCH EXPERIENCE

Research Assistant, NAIST Apr 2026 - Present
Cybernetics and Reality Engineering Lab | Advisor: Prof. Kiyoshi Kiyokawa

Research in human-computer interaction and VR/XR.

Research Visit, Colorado State University Jan - Jul 2026
Natural User Interaction Lab | Host: Prof. Francisco R. Ortega

Completed an on-site research visit in Fort Collins, Colorado.

Master's Thesis Research, NAIST Aug 2024 - Mar 2026
Cybernetics and Reality Engineering Lab | Advisor: Prof. Kiyoshi Kiyokawa

- Investigated perceived resistance and task load in head-following AR windows, comparing Head-Locked and Tag-Along interfaces across follow-up delays and window sizes.
- Independently implemented the system and designed, conducted, and analyzed a 24-participant within-subject study; wrote the manuscript.
- Analyzed resistance ratings, NASA-TLX scores, and gaze-head metrics using aligned rank transform ANOVA; first-authored the IEEE TVCG paper [1].

RESEARCH EXPERIENCE (CONTINUED)

Tactile Sensing Glove Research, The University of Aizu May 2023 - May 2024

Undergraduate research and follow-up work | Advisor: Prof. Lei Jing

- Built and evaluated a 46-sensor tactile glove for recognizing 10 Japanese fingerspelling signs, using data from five participants.
- Handled device assembly, data collection, recognition-model implementation, analysis, and manuscript writing; PCB design and fabrication were carried out by collaborators.
- Compared LSTM and k-NN with Dynamic Time Warping; the latter achieved 87% mean accuracy with group 5-fold cross-validation. First-authored the MCSoC 2024 paper [2].

SELECTED PROJECT

ChattyPlanter | Kansai Tech Creator Challenge 2024

Project advisor: Prof. Ismail Arai

- Developed a working plant-mediated conversation prototype integrating environmental sensing and GPT-based dialogue; independently handled concept and interaction design, implementation, prototyping, and presentation.

TEACHING & MENTORING

Teaching Assistant, NAIST Research Internship Apr 2025 | 4 days

- Mentored one fourth-year undergraduate in Unity/C# implementation of desktop pseudo-haptics, adjusting shaders and virtual-cloth motion to convey contact sensations; guided development of a working demonstration.

Teaching Assistant, NAIST Research Internship 17-21 Feb 2025

- Taught one third-year undergraduate Unity, Bambu Lab 3D printing, device assembly, and implementation from the basics; guided a working PaCaPa-based prototype for impact feedback during virtual contact.

SCHOLARSHIPS & RESEARCH SUPPORT

JASSO Student Exchange Support Program Mar - Jul 2026

Scholarship for study abroad under an institutional agreement: JPY 110,000/month for five months (JPY 550,000 total), supporting the CSU research visit.

Overseas Research Support, NAIST J-PEAKS 2026

JPY 320,000 in support for the CSU research visit.

Kansai Tech Creator Challenge (K-TCC) 2024

Selected project; JPY 550,000 in project funding received for ChattyPlanter.

TECHNICAL & RESEARCH SKILLS

Programming: Python, C#, R. **Development:** Unity, RStudio.

Research methods: User-study design and execution; statistical analysis of experimental data; aligned rank transform ANOVA; data visualization.

Libraries: NumPy, pandas, Matplotlib, scikit-learn, TensorFlow.